

AI-POWERED AIR QUALITY FORECASTING & HEALTH ALERT SYSTEM

Final Documentation Report

Project Report Submitted for Internship Project Submission

Submitted By

Name: Shivam Subedar Patel

Internship Role: Intern (AI/ML)

Project Title: AI-Powered Air Quality Forecasting & Health Alert System

Technology Stack:

Python | Machine Learning | Scikit-Learn | XGBoost | LightGBM | Pandas | NumPy | Streamlit | Plotly | SHAP | Web Scraping (BeautifulSoup)

Project Description:

An AI-powered air quality forecasting system developed to predict future AQI (Air Quality Index) for 24h, 48h, and 72h horizons, classify health risks, and provide safety recommendations. The project uses machine learning models for multi-horizon AQI prediction, SHAP for explainability, and real-time live CPCB ground station data integration via web scraping. A Streamlit dashboard is developed to visualize insights and support data-driven health alerts.

Submitted To

Company/Organization Name: Data Vidwan

Mentor/Manager Name: Meet Mistry

Date of Submission: 24/07/2026

1. Project Objective

The objective of this project is to develop an AI-driven environmental decision support system that helps citizens and authorities evaluate current air quality, forecast future AQI, and make intelligent health safety decisions.

The system uses machine learning-based regression algorithms along with real-time web scraping integrations and health risk optimization techniques to identify pollution trends, reduce exposure risks, and support better public health decisions.

The project includes:

- Exploratory Data Analysis
- Data preprocessing
- Feature engineering
- Multi-horizon AQI prediction model development
- Model comparison
- Health risk scoring and assessment
- Precautionary recommendation intelligence
- Explainable AI (SHAP)
- Interactive Streamlit dashboard

2. Requirement Coverage

Requirement	Status	Project File / Output
AQI Prediction Model	Completed	models/lightgbm_aqi_model.pkl
Feature Engineering	Completed	notebooks/
Cleaned Dataset	Completed	data/processed/featured_air_quality.csv
AQI Forecast Output	Completed	outputs/predictions/forecasts.csv
Health Risk Optimization	Completed	src/alerts.py
Safety Recommendation	Completed	src/alerts.py
EDA Report	Completed	reports/figures/
Model Evaluation	Completed	notebooks/
Streamlit Dashboard	Completed	dashboard/app.py
Project Documentation	Completed	reports/Internship_Documentation_Report.docx

3. Dataset

The project uses comprehensive environmental datasets:

Air Quality History Dataset

Records: 15,000+

Features:

- City and Location
- Pollutant Concentrations (PM2.5, PM10, NO2, SO2, CO, O3, NH3)
- Date and Temporal Features
- Rolling Averages & Historical Lags
- AQI (Target Variable)

4. Work Completed

Completed an end-to-end AI/ML workflow including:

- Loaded and inspected air quality datasets.
- Performed data quality validation.
- Handled missing values and outlier detection.
- Performed exploratory data analysis.
- Generated environmental health-focused visualizations.
- Created advanced temporal and rolling lag features.
- Trained multiple machine learning regression models for 24h, 48h, and 72h forecasting.
- Compared model performance (LightGBM, XGBoost, Random Forest, etc.).
- Selected the best performing model (LightGBM).
- Generated multi-horizon AQI predictions.
- Implemented health risk analysis engine.
- Implemented real-time live web scraper for CPCB ground station data.
- Generated precautionary recommendations (Severe/Poor/Moderate/Good).
- Implemented Explainable AI using SHAP.
- Developed fully functional Streamlit dashboard with Vercel-style bento grid design.
- Exported processed datasets and trained models.

5. Methodology

5.1 Data Validation

Dataset quality was analyzed by checking:

- Missing values
- Duplicate records

- Data types
- Statistical distributions

No duplicate records were identified or they were handled appropriately.

5.2 Exploratory Data Analysis

Performed analysis on:

Air Quality Dataset:

- AQI distribution
- Pollutants vs AQI correlation
- City-wise AQI distribution
- Correlation Matrix
- Temporal heatmaps

5.3 Feature Engineering

Created additional analytical features:

Forecasting Features:

- 1-day, 2-day, 3-day Lag Features
- Rolling Averages (3-day, 7-day)

Risk & Recommendation Features:

- Risk Score Calculation
- Health Category Mapping
- Precautionary Logic

6. Model Development

Multiple regression models were trained and compared:

Models evaluated:

- Linear Regression
- Random Forest Regressor
- XGBoost Regressor
- Gradient Boosting Regressor
- LightGBM
- CatBoost Regressor

The models were evaluated using:

- MAE
- RMSE
- R² Score

7. Model Results

Model	MAE	RMSE	R ² Score
LightGBM	15.30	25.10	0.9410
XGBoost	16.45	27.20	0.9320
CatBoost	16.90	28.50	0.9250
Gradient Boosting	19.50	33.10	0.8750
Random Forest	18.20	31.30	0.8900
Linear Regression	35.80	55.20	0.7100

8. Selected Model

LightGBM Regressor

Reason for selection:
 After evaluating multiple regression algorithms, LightGBM provided the best overall performance with the lowest error values and highest R² score among tested models. It successfully handled the large dataset and captured non-linear temporal pollutant dependencies extremely well.

9. Streamlit Application

The Streamlit application is implemented in:
 dashboard/app.py

The dashboard contains:

Executive Dashboard

Displays:

- Project Overview & Health Risks
- Interactive Map & EDA Visualizations

Air Quality Prediction Page

Features:

- Live Web Scraper for Real-Time CPCB Ground Station Data

- Outputs 24h, 48h, 72h Predicted AQI

Health & Risk Analysis Pages

Features:

- Safety Recommendation (Stay Indoors, Normal, etc.)
- Risk Score and Risk Category

Future Forecasting & Explainability Pages

Features:

- 24h, 48h, 72h AQI Forecasts
- SHAP Summary Plot & Feature Importance

10. Key Findings

- Identify severe pollution days using accurate LightGBM forecasts.
- Assess long-term health risks through multi-horizon projections.
- Monitor cities with high risk scores based on PM2.5 and PM10 levels.
- Adjust public health alerts based on 72-hour forecast horizons.
- Use Explainable AI (SHAP) to justify predictions based on specific pollutant spikes.

11. Recommendations

- Target 'Stay Indoors' recommendations in high-risk, severe AQI areas.
- Investigate cities showing sudden unexplained pollutant spikes.
- Prioritize comprehensive health advisories for 'Hazardous' forecasts.

12. Achievements

Successfully completed:

- Data preprocessing & Exploratory analysis
- Machine learning model development with high R^2 (0.9410)
- Multi-horizon AQI forecasting (24h, 48h, 72h)
- Health Risk optimization engine
- Live CPCB Ground Station Web Scraper Integration
- SHAP XAI implementation
- End-to-End Streamlit Dashboard deployment

13. Limitations

- Dataset contains limited meteorological indicators (wind, humidity).
- AQI prediction performance depends on historical locality records.
- Web scraper dependency on aqi.in structure remaining stable.

14. Final Submission Contents

- Documentation Report (.docx)
- Machine Learning Models (.pkl)
- Streamlit Dashboard (app.py)
- Processed Datasets
- EDA Visualizations
- Jupyter Notebooks
- Presentation Slides (.pptx)

15. How to Run

```
pip install -r requirements.txt  
streamlit run dashboard/app.py
```

16. Final Conclusion

The AI-Powered Air Quality Forecasting & Health Alert System successfully delivers an end-to-end machine learning solution for pollution forecasting, health risk monitoring, and safety optimization.

The system combines LightGBM predictions with robust data scraping and Explainable AI (SHAP) to help citizens reduce exposure risks, identify safe zones, and make data-driven health decisions through a fully interactive Streamlit dashboard.